

Day 13

Content modifier variable

\* Standard Expressions:

Current time stamp (in UTC): ✓

Current time stamp (in IST): \$ {date-with-timezone:  
now:IST: - - -

\* msgid ? → expression: \$ {SOP\_MPLID?}  
property ID.

\* iflow name: → \$ {camelID}  
↳ small letter

→ Requirement → When ever we processing the message

We need process the "previous message" ✓

eg: processed empID ① → before msg, what is that?

→ what is a message → What ever the data

you are sending from Source System to target

System → we called it as a message

(or) payload ✓

\* When we are talking about SAP CPI  
perspective then we need to remember ✓  
that → with the incoming payload → we  
are sending → message headers & message  
properties. → So that's why we called it  
as a message (properties, headers, and body) ✓

→ Requirement → for current message processed

→ we need to fetch the previous message  
details also? now what you will do?

→ Can we store the previous message details?  
or let say 1st input Emp①, next time 2nd

Emp② → now for 2 output we need

to get the 1st msg details also →

is there a way where can we use it here?

→ if we create a variable in exchange property  
and store previous msg details and call it  
in current msg output → does it work →?

Yes or NO ✓

Because  $\rightarrow$  cpi  $\rightarrow$  does not store the  
data ? then how you will do ?

\* The important point is the variables which  
are created at message header or, exchange  
property or body  $\rightarrow$  Those will Reach  
to the End message and deleted.  
by message header values and body will  
be sent to the Target System

\* Now for the above Requirement  $\rightarrow$   
To provide the previous msg details in the  
current details of msg  $\rightarrow$  then we will  
use the Source types in Content modifier

are  $\rightarrow$  local variable & Global variable

\* How can I use this local variable &  
Global variable.

\* Now about the local variable?

→ The point here is why that msg is not local variable getting deleted at End message?

Ans: properties, header, payload → created → iflow

and local variable & Global variables are created outside the iflow

\* Body(Message) → old message details:  
→ current message details:

→ message Body:

Hi

Welcome to SAP CP1 class

Please find the below details

Current message details

iflow name:

msgID

Current Timestamp (in UTC)

Current Timestamp (in IST)

EmpID:

Old message Details :

Key's are same as above message details

But the

values should be change

So for the above Scenario → local variable

Concept? How to do that?

- ① how to write the data to local variable
- ② how to read the data to local variable
- ③ Where the local variable stored

① → ~~Ans~~ → Persistence Tab (Palette) → write variables

↓  
To write the data to the variable

if we want to write the data to any variable then we will use write variable

Palette which is located in persistence Tab

→ Requirement → Write the previous msg → ~~local~~ local variable

iflow name will be same for the all the messages

So we will ignore the iflow name (because it is common)

\* important thing is msgid → every msg → New ID → will be generated (new)

→ here → get the msgid (previous) → store in variable

\* Same whatever we will do in Content modifier We will do it in Body as well → let's see

① → goto → write variable → Add → oldmsgid name (key)

→ Type (property) → SAP\_messageProcessing logid (Value)

② Timestamp → When was that msg ran ~ Both in UTC & IST and the previous

msg that employee ID details

→ So we will create multiple variable here

Previous msgDate (in UTC) ✓  
" " in DIST ✓  
" ~~msgid~~ EmpID ✓  
" " msgid ✓

} We need to use the standard expression

So here I am storing the data in a local  
Variable → We need to use write variable  
function (Persistence Tab)

→ ② How are you going to read the data  
from that local variable (above created)

→ goto → Content modifier (earlier created) →  
exchange property → create all the fields

Source type → local variable

eg: 9 AM → 1 msg

10 AM → 1 msg

11 AM → 1 msg

With this msg →  
how you will get  
previous msg  
details

\* So the flow will be whenever we hit the sender request from the postman it will store those required details in write variables and we will call those variables in exproperty → message Body

postman → iflow → write variables → Content  
modifier → Body

This is the flow  
So from starting point to End like this  
it will follow

② Where ~~are~~ the variables getting stored  
→ monitor → manage stores → variables (here those variables will be stored)

\* Default value → N/A ✓

① → what happens if we use the write variable before the Content modifier

ABLS: NO ? flow will be complex.

→ first we will check whether the data is available or not ✓

\* → What is the difference between local variable and global variable

local → means → within that flow ✓

Global → means → out side that flow  
(tenant level)

\* Global Scope (option) → write variable

→ looking at the visibility in monitor tab  
we can clearly say whether that variable  
is local or global ✓

→ global variable → demo →

\* Real-time Scenario → Where do we use the local variable and global variable

① → Scenario.

→ let's say → Server → username, password → Basic Auth

② → Scenario

→ let's say → Server → Auth → token based (OAuth2)

Token → 567 ✓  
(value)

Global variable

Some flows need this token which is acts as a global variable at that time  
We will use this concept ✓

→ local variable

We want → Send → data → Two type of option

→ full load (or) Delta load

↓  
filter

with  
out  
any  
issue

Suppose a table  
has '10' records

Read the  
complete data

Scenario → Particular  
time updated  
records

Delta load: Condition if current timestamp < previous msg timestamp

Important points →

Format for declaring variables (write variable)

'(' ')' → not allowed

no → Spaces

like (in<sub>x</sub>IST)X inIST ~

↳ space also not allowed ✓

① Ask to param -?

XSLT (mapping) ~  
XML → XSLT → XML  
↓

About ① Process Direct Adapter

② AMQP (Queues)

③ JDBC Adapter

④ IDOC Adapter